

GREENPASS

Digital sustainability infrastructure for African small and medium-sized businesses.

PROJECT STAGE
Prototype

SECTOR
Digital Technology in Service of the Green Economy

PROJECT BRIEF

GreenPass helps businesses measure their environmental performance, identify practical ways to improve it, and present credible sustainability information to the people they want to do business with.

Prepared for project competition submission

1. The Problem

A large number of small businesses operate with limited information about their energy use, fuel consumption, waste, transport activity and related environmental impact. The information may exist in receipts, meter readings, fuel records and everyday business operations, but it is rarely brought together in one place.

This creates two problems. First, businesses miss opportunities to reduce operating costs because they cannot easily see where resources are being wasted. Second, they struggle to provide organised sustainability information when a bank, investor, corporate buyer or supply-chain partner asks for it.

The gap is not simply a lack of environmental awareness. It is a lack of simple digital infrastructure that turns ordinary business records into useful sustainability information.

Why this matters now

Sustainability reporting and ESG requirements are becoming more relevant to Nigerian businesses and financial institutions. At the same time, businesses participating in larger supply chains increasingly need better records of how they operate. GreenPass is designed to help smaller businesses prepare for that shift without requiring them to become environmental specialists.

Operational visibility

Businesses can see where energy, fuel, transport and waste are affecting their operations.

Better business decisions

The platform connects environmental performance with practical actions, expected savings and improvement priorities.

Business credibility

A structured sustainability profile gives SMEs a clearer way to communicate their progress.

Green-finance readiness

Organised sustainability information can make it easier for businesses to respond to environmental information requirements from finance and investment partners.

2. The GreenPass Solution

GreenPass is a web-based platform that turns simple operational information into a clear sustainability profile for an SME.

A business does not need specialist environmental knowledge to use it. The onboarding process collects information that the business can reasonably access, such as electricity use, fuel consumption, transport activity, waste and production data.

The platform then produces four practical outputs:

Green Business Score

A simple score that summarises the business's current environmental performance across selected indicators.

Environmental Profile

A breakdown of key activities such as energy, fuel, transport and waste, presented in a business-friendly dashboard.

Improvement Roadmap

Prioritised actions showing what the business can improve, the likely cost level, potential savings and environmental impact.

Green Business Passport

A digital sustainability profile with a unique verification link and QR code that can be shared with business partners.

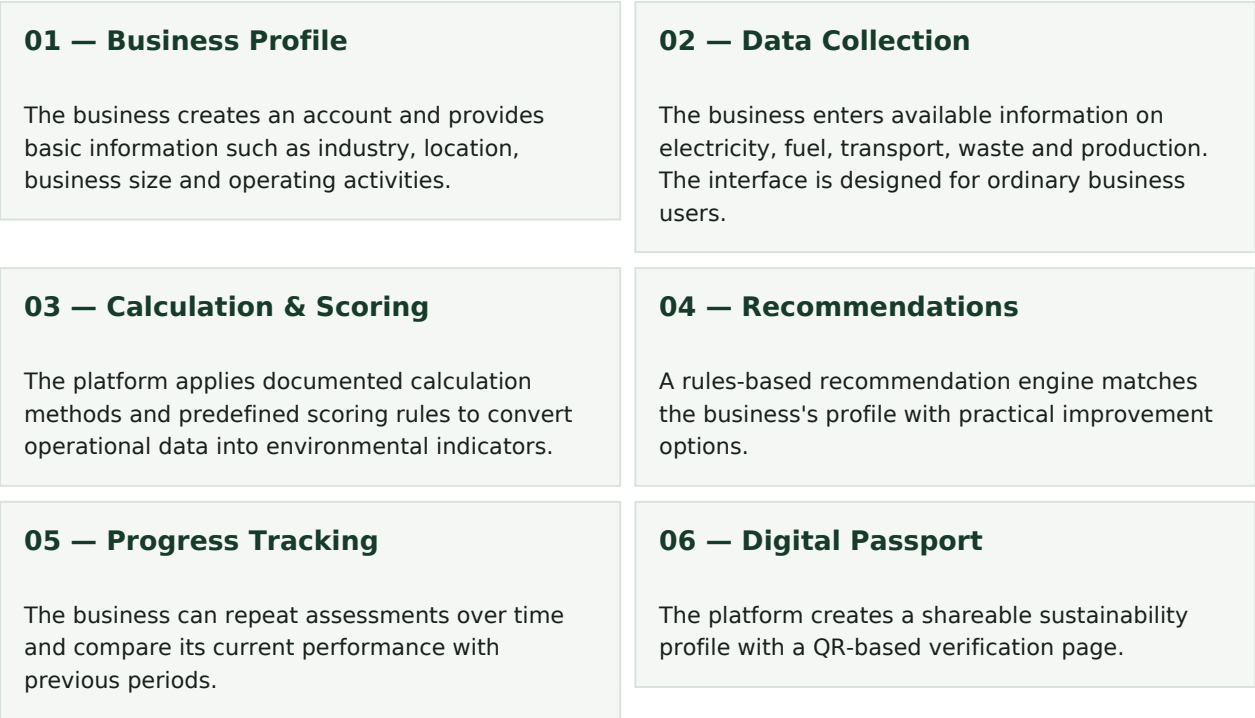
The product is designed around one simple progression:

MEASURE → UNDERSTAND → IMPROVE → PROVE

The purpose is not to give a business another complicated report. The purpose is to turn information into action and then document the improvement.

3. How the Platform Works

GreenPass can be built and demonstrated as a working web application at prototype stage.



Prototype architecture

- Web application — business accounts, forms and dashboards
- Calculation engine — structured environmental calculations and scoring
- Recommendation engine — rule-based matching of actions to business conditions
- Database — operational records and historical performance
- Reporting layer — downloadable reports and digital passport
- Verification layer — unique business IDs and QR-linked public verification

4. Example Business Journey

The following example illustrates how the prototype would work. Figures shown here are illustrative and are not presented as measured results.

BUSINESS	INPUT	GREENPASS OUTPUT
Food processing SME	Electricity, diesel, delivery kilometres, waste and production volume	Environmental profile + Green Business Score + improvement priorities
Operational finding	High diesel use during daily operations	Flagged as a major improvement opportunity
Recommendation	Assess solar/hybrid energy options and reduce unnecessary generator runtime	Estimated cost, potential savings and environmental impact are shown for comparison
Progress	Business repeats the assessment after implementation	GreenPass records the change and updates the business profile
Proof	Business chooses to share its sustainability profile	Green Business Passport + QR verification page

Important design principle

GreenPass should never present an estimate as a certified measurement. Calculations and scores should be transparent about their data source, assumptions and level of verification. This keeps the product useful without making claims the prototype cannot support.

5. Business Model

GreenPass is designed as a software-as-a-service business with three main customer levels.

Free

Basic assessment, initial Green Business Score and a simple environmental profile. The free tier lowers the barrier to adoption.

Professional

Advanced tracking, detailed improvement roadmaps, downloadable reports, historical comparisons and the Green Business Passport.

Enterprise

For banks, large companies, investors and supply-chain programmes that need to assess or monitor multiple SMEs.

Additional services

Optional verification, reporting support, API integration and sustainability data services can provide additional revenue.

Who pays?

The SME is the first paying customer for professional features. Larger organisations become higher-value customers when they need information from many SMEs rather than a single business. This creates two connected markets: businesses that need to improve and organisations that need reliable information about businesses.

The long-term opportunity is to become a trusted digital layer between SMEs and organisations that care about environmental performance, financing and supply-chain requirements.

6. Feasibility & Scalability

The first version does not depend on expensive physical infrastructure. It can be built as a standard web platform using established technologies and a structured set of calculation and recommendation rules.

The prototype can focus on a limited number of sectors and indicators rather than attempting to solve every sustainability problem at once. This makes the first release practical for a solo developer while leaving room for later expansion.

Core prototype components

User & Business Management Registration, business profiles, assessment history and account management.	Assessment Engine Structured forms and calculation logic for selected environmental indicators.
Dashboard Charts, scores, trends and category breakdowns that a business owner can understand.	Recommendation Engine Rules that match business conditions with practical improvement actions.
Reporting Downloadable business reports and sustainability summaries.	Verification Unique GreenPass IDs and QR-linked public profiles.

Scaling strategy

- Phase 1: Nigeria — selected SME sectors and pilot businesses.
- Phase 2: Expand sector coverage and strengthen partnerships with finance and corporate supply chains.
- Phase 3: Adapt the platform to other African markets by adding local data, benchmarks and relevant calculation factors.

7. Green Economy Impact

GreenPass contributes to the green economy by making sustainability information useful at the level where millions of everyday business decisions are made.

Lower resource waste

Businesses can identify inefficient energy, fuel, transport and waste practices and prioritise improvements.

Better environmental decisions

The platform turns scattered operational information into indicators that support practical decisions.

Stronger SME participation

Smaller businesses gain access to sustainability tools that are normally designed for larger organisations.

Better information for finance

Structured environmental information can support future green-finance and sustainability-linked business processes.

More transparent supply chains

Businesses can share a consistent sustainability profile with larger buyers and partners.

Continuous improvement

Historical tracking encourages businesses to measure progress rather than treating sustainability as a one-time exercise.

Impact will be measured through indicators such as:

- Number of SMEs assessed
- Number of businesses actively tracking performance
- Reported reduction in energy or fuel consumption
- Reported waste reduction or recovery
- Number of businesses completing improvement actions
- Number of verified Green Business Passports issued

8. Competitive Position

GreenPass is not positioned as a replacement for professional environmental consultants, auditors or formal certification bodies. It is a digital entry point that helps SMEs organise their information and improve their readiness before they need more specialised services.

Its differentiation is the combination of measurement, improvement planning, progress tracking and a shareable business sustainability profile in one SME-focused platform.

Alternative	Typical strength	Gap GreenPass addresses
Spreadsheets	Flexible and familiar	Manual calculations, weak tracking and no structured business profile
Generic carbon calculators	Quick footprint estimate	Limited connection between measurement, business improvement and verification
Large ESG platforms	Advanced enterprise reporting	Often too complex or expensive for smaller businesses
Consultants	Specialist expertise	Higher cost and less accessible for routine SME tracking
GreenPass	SME-focused digital workflow	Measure → improve → track → share

The strongest part of the product is not the calculation formula. It is the workflow around the data: helping the business act on it, track the change and communicate the result.

9. Prototype Roadmap

The development plan is intentionally focused. The first objective is not to build a giant platform; it is to prove that a Nigerian SME can complete an assessment, understand the result, take an action and return to measure progress.

STAGE	FOCUS	OUTPUT
1. Prototype	Core platform, business onboarding, assessment forms, scoring and dashboard	Working GreenPass prototype
2. Pilot	Test with selected Nigerian SMEs and collect feedback	Validated workflow and improved product
3. Improve	Refine recommendations, reporting and verification	More reliable and easier-to-use platform
4. Launch	Introduce free and paid plans	Commercial Nigerian product
5. Scale	Partnerships and expansion into additional African markets	Regional sustainability-data platform

What support would unlock

The most valuable early support would be access to pilot SMEs, technical and environmental guidance, mentorship, and introductions to financial institutions or organisations working with SME sustainability and green finance.

10. The Vision

GreenPass starts with a simple question:

“Can a small business understand and improve its environmental performance without needing a specialist?”

The long-term goal is bigger than a calculator or reporting tool. GreenPass aims to make sustainability information part of everyday SME decision-making and give businesses a simple digital identity for communicating their environmental progress.

As the platform grows, that information can become useful to more parts of the economy: businesses, buyers, financial institutions, investors and supply-chain programmes.

GREENPASS

**Measure what your business uses.
Understand where you can improve.
Track the change.
Show the progress.**

Prototype status

GreenPass is currently at prototype stage. The next step is to turn the concept into a working pilot, test the workflow with real businesses, improve the calculations and recommendations, and establish a reliable approach to verification.

Supporting document prepared for competition submission.